Complete Guide: Creating a Login Database in phpMyAdmin (XAMPP)

This guide will walk you through creating a database and table for user login functionality using phpMyAdmin in XAMPP.

Prerequisites

- XAMPP installed and running
- Apache and MySQL started in XAMPP Control Panel

Step-by-Step Instructions

Step 1: Access phpMyAdmin

1. Make sure **Apache** and **MySQL** are running (green status) in XAMPP Control Panel
2. Open your web browser
3. Go to: http://localhost/phpmyadmin
4. You should see the phpMyAdmin dashboard

Step 2: Navigate to SQL Tab

1. In the phpMyAdmin interface, click on the **"SQL"** tab in the top navigation menu
2. This is where you'll paste and execute your SQL code

Step 3: Execute the Database Creation Code

1. **Copy and paste this entire code block** into the SQL text area:

```
sql
CREATE DATABASE IF NOT EXISTS usersdata;
USE usersdata;

CREATE TABLE IF NOT EXISTS users (
  id INT AUTO_INCREMENT PRIMARY KEY,
  name VARCHAR(100) NOT NULL,
  email VARCHAR(255) UNIQUE NOT NULL,
  password VARCHAR(255) NOT NULL,
  role ENUM('user', 'admin') DEFAULT 'user',
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP
);
```

2. Click the **"Go"** button to execute the SQL commands

Step 4: Verify Database Creation

1. Look at the left sidebar in phpMyAdmin
2. You should see a new database called usersdata appear in the list
3. Click on usersdata to select it
4. You should see the users table listed

Step 5: Verify Table Structure

1. Click on the users table
2. Click the **"Structure"** tab to view the table columns
3. Verify you have these columns:
    - id (primary key, auto-increment)
    - name (VARCHAR 100)
    - email (VARCHAR 255, unique)
    - password (VARCHAR 255)
    - role (ENUM: user/admin)
    - created_at (TIMESTAMP)
    - updated_at (TIMESTAMP)

Understanding Your Database Structure

What You Just Created:

- **Database**: usersdata - stores all user information
- **Table**: users - contains individual user records

Table Columns Explained:

- id: Unique identifier for each user (automatically numbers users 1, 2, 3...)
- name: User's full name (required field)
- email: User's email address (must be unique for each user)
- password: Will store encrypted passwords (always encrypt passwords!)
- role: User permission level ('user' or 'admin')
- created_at: Automatically records when user registered
- updated_at: Automatically updates when user info changes

Adding Test Data (Optional)

Step 6: Insert Test Users

1. Go to the **SQL** tab again

2. Paste this code to add sample users:

sql

```
USE usersdata;


-- Insert a regular user

INSERT INTO users (name, email, password, role)

VALUES ('John Doe', 'john@example.com',
'$2y$10$92IXUNpkjO0rOQ5byMi.Ye4oKoEa3Ro9llC/.og/at2.uheWG/igi', 'user');


-- Insert an admin user

INSERT INTO users (name, email, password, role)

VALUES ('Admin User', 'admin@example.com',
'$2y$10$92IXUNpkjO0rOQ5byMi.Ye4oKoEa3Ro9llC/.og/at2.uheWG/igi', 'admin');
```

Note: The passwords are hashed versions of "password" - in a real application, you should use proper password hashing.

Step 7: View Your Data

1. Click on the users table
2. Click the **"Browse"** tab
3. You should see the test users you just added

Connecting to Your Database from PHP

Here's a basic PHP code snippet to connect to your database:

php

```
<?php

$servername = "localhost";

$username = "root";

$password = ""; // Default XAMPP password is empty

$dbname = "usersdata";


// Create connection

$conn = new mysqli($servername, $username, $password, $dbname);


// Check connection
```

```
if ($conn->connect_error) {
    die("Connection failed: " . $conn->connect_error);
}
echo "Connected successfully to usersdata database!";
?>
```

Troubleshooting Common Issues

If you get errors:

- **"No database selected"**: Make sure you executed both CREATE DATABASE and USE usersdata commands
- **Access denied**: Check that MySQL is running in XAMPP
- **Table already exists**: The IF NOT EXISTS clause should prevent this, but if needed, you can drop the table first

To Start Over:

1. In phpMyAdmin left sidebar, click on usersdata database
2. Click **"Operations"** tab
3. Scroll down and click **"Drop the database (DROP)"**
4. Confirm, then run the SQL code again

Security Notes

- Always hash passwords using password_hash() in PHP
- Never store plain text passwords
- In production, change the default MySQL root password
- Use prepared statements to prevent SQL injection